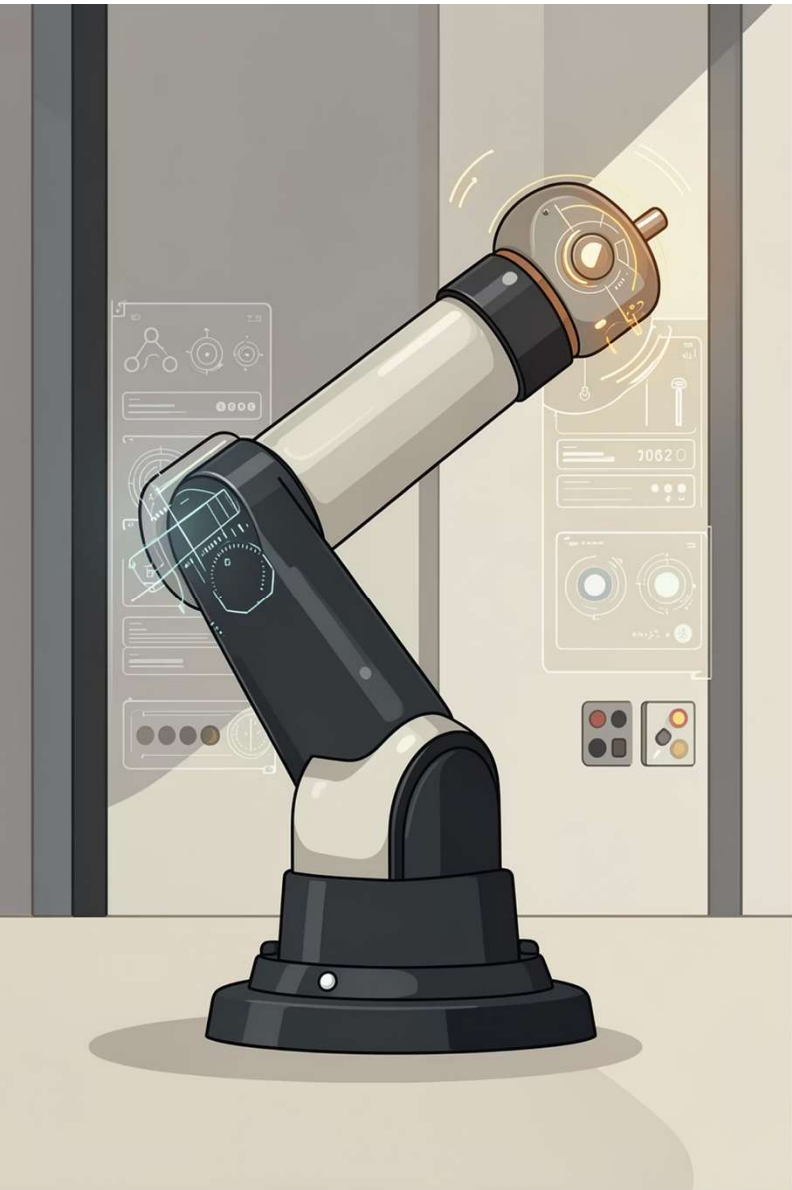




ROBOT ABSTRACTION

A developer library that turns natural language into typed, policy-guarded robot capabilities — across simulation, ROS 2, and selected real hardware.

1

GEMINI

2

WEBOTS

3

SIMULATION-FIRST

4

OPEN SOURCE

ROBOTS ARE SCALING. CONTROL IS STILL FRAGMENTED.



Robotics adoption is accelerating across industry, logistics, service robotics, education, and research. LLMs can now convert human instructions into structured tool calls.

But the stack is still fractured – ROS 2, simulators, hardware SDKs, and custom APIs don't speak the same language.

i The opportunity: A safe abstraction layer that moves from natural language to robot action – without rebuilding glue code every time.

"LLM SAYS MOVE" IS NOT ENOUGH

Robots are physical systems. Raw LLM output cannot be trusted to directly command hardware.

FRAGMENTED APIS

Every robot platform exposes a different interface – no shared abstraction exists.

SIM VS. HARDWARE GAP

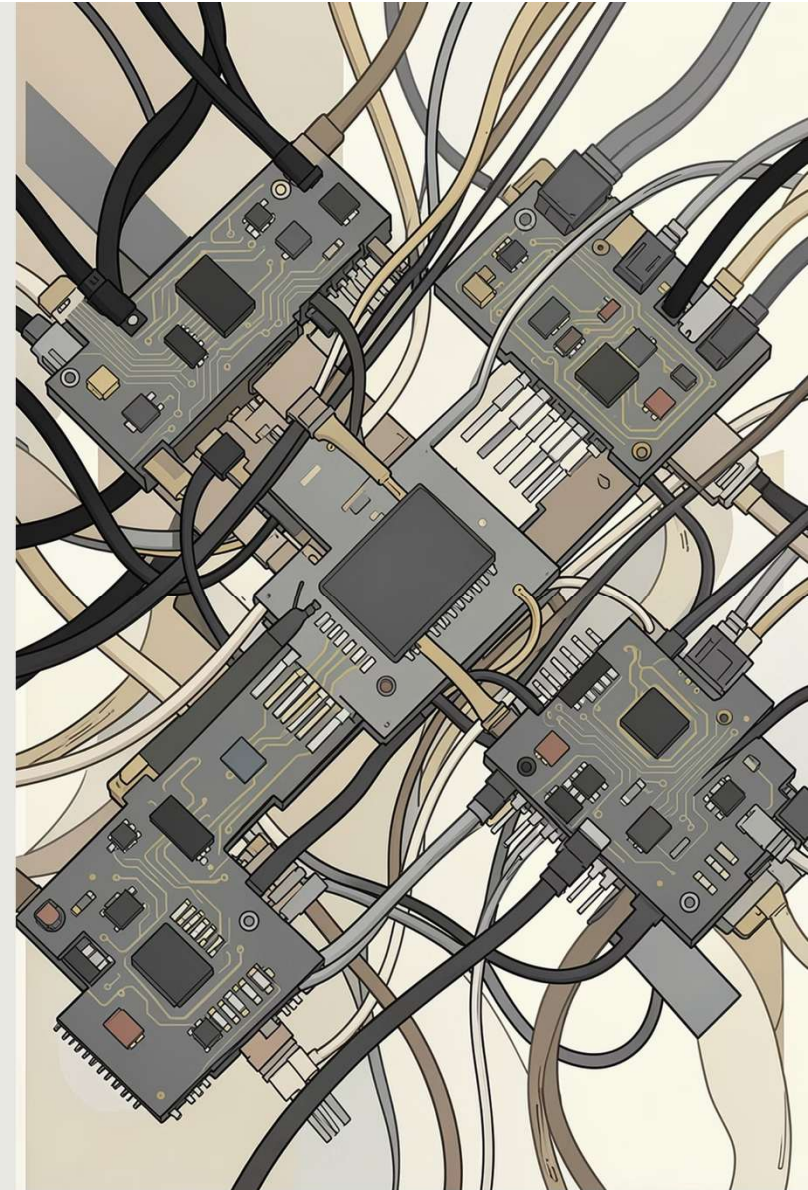
Simulators and real hardware behave differently; parity is rarely guaranteed.

UNSAFE LLM OUTPUTS

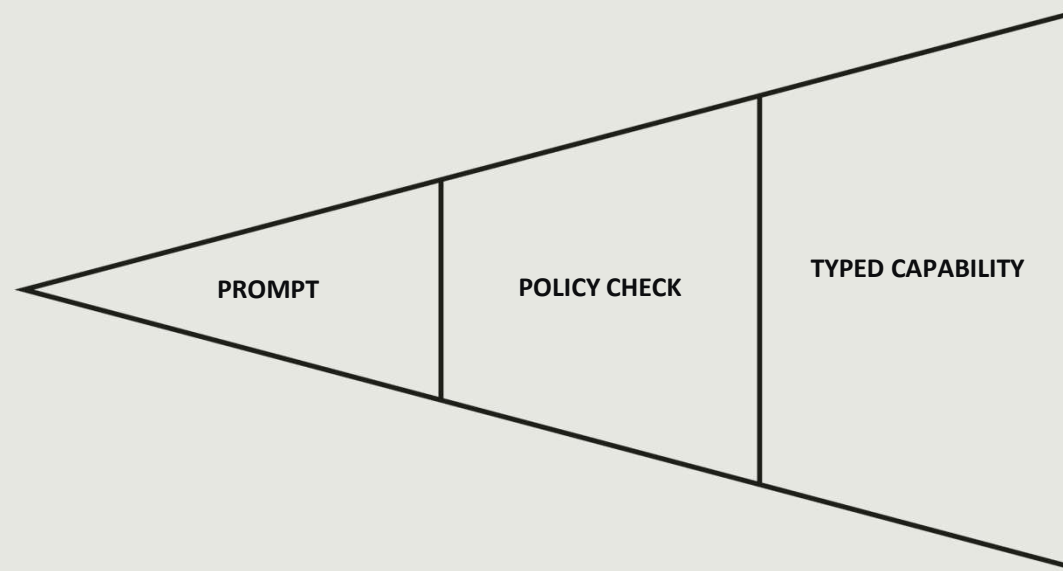
Models may hallucinate unsupported or dangerous actions with no built-in guardrails.

NO SAFETY PRIMITIVES

Teams hand-roll logs, dry runs, safety gates, and e-stop logic from scratch every time.



ONE SEMANTIC CONTROL LAYER FOR ROBOTS



The LLM proposes actions. The runtime validates and dispatches them safely – the model never holds direct hardware authority.

NAVIGATE

Move to waypoint

DOCK

Return to station

INSPECT OBJECT

Targeted analysis

STOP / ABORT

Emergency halt

CAPTURE IMAGE

Vision snapshot

QUERY STATE

Robot telemetry

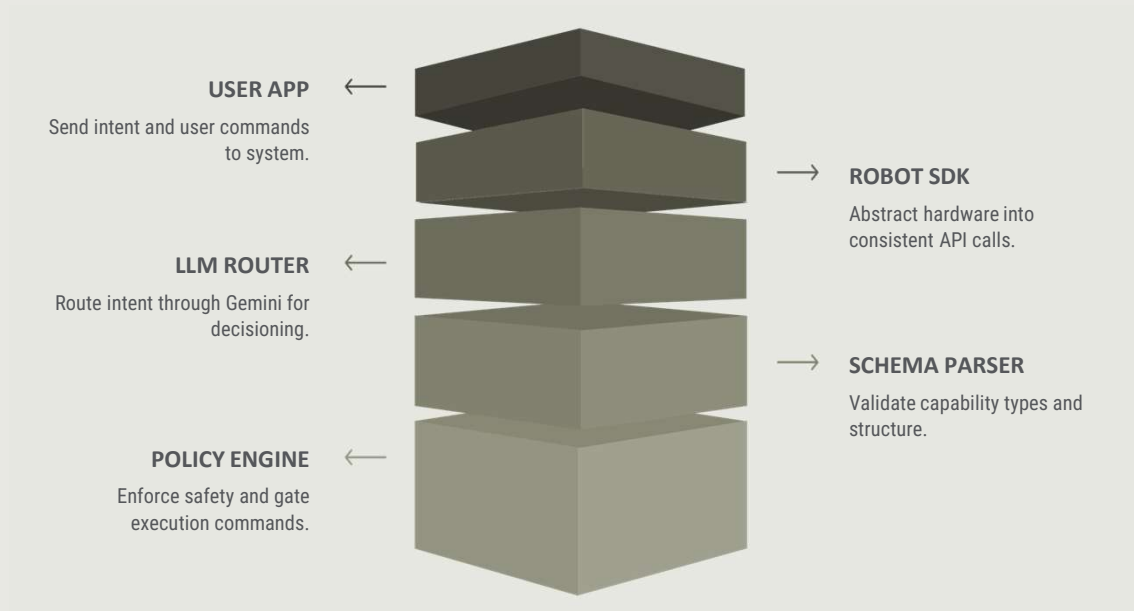
HOW ROBOT ABSTRACTION WORKS

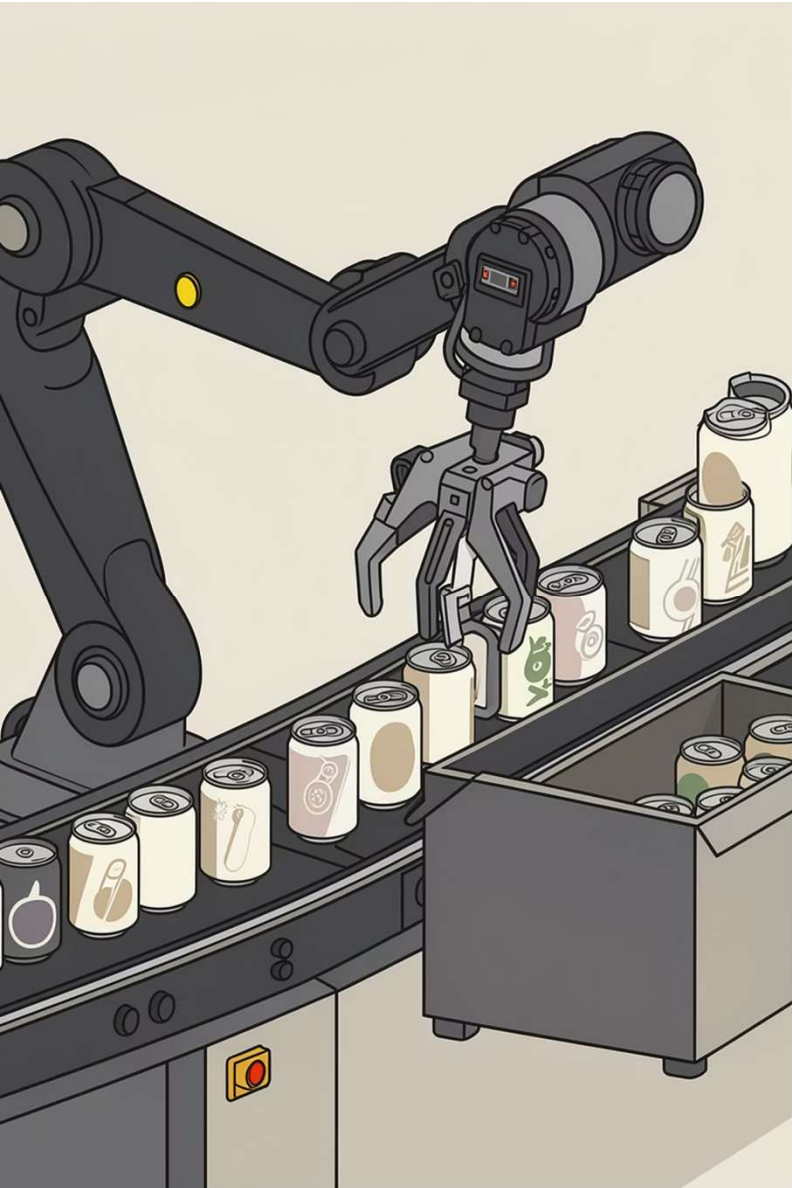
LAYERED BY DESIGN

Each layer has a single responsibility. The LLM router handles intent; the schema parser enforces types; the policy engine gates execution.

 **Core principle:** The model never gets direct hardware authority. Every action is validated before dispatch.

- Gemini-first routing, multi-LLM ready
- Adapters for ROS 2, Webots, Gazebo, TurtleBot3
- Full telemetry, trace, and replay support





MVP DEMO: PICK 5 CANS

COMMAND BREAKDOWN

"Pick 5 cans from the conveyor and place them in the bin."

01

IDENTIFY CANS

02

GRASP CAN

03

MOVE TO BIN

04

RELEASE ITEM

05

REPEAT UNTIL 5

SAFETY GATE CHECKS

- ✓ Is the gripper force within safe limits?
- ✓ Is collision detection active around the conveyor?
- ✓ Does the bin have enough capacity?
- ✓ Can the robot stop safely between picks?

✓ **Demo highlight:** The robot picks approved cans and fruit, places them accurately in the bin, and blocks unsafe motions with a trace log.

CUSTOMER SUCCESS STORY



10,000

CANS PER DAY

10,000

FRUITS PER HOUR

SUPPORTED ROBOT PLATFORMS



FactoryFlowCopilot



FactoryFlowCopilot helps Karmar Packaging orchestrate flexible picking and packing operations with reliable, production-ready automation.

Made with GAMMA



MVP DEMO: INSPECT AISLE B

COMMAND BREAKDOWN

"Inspect aisle B and report obstacles."

01

NAVIGATE TO AISLE B

02

CAPTURE IMAGE

03

QUERY ROBOT STATE

04

REPORT RESULT

SAFETY GATE CHECKS

- ✓ Is this capability allowed?
- ✓ Simulation or hardware mode?
- ✓ Target zone permitted?
- ✓ Can the action be aborted mid-flight?

✓ **Demo highlight:** One valid command executes. One unsafe command is blocked – visibly, with a trace log.

WHO NEEDS THIS FIRST?



ROBOTICS DEVELOPERS

Prototype LLM-controlled robot workflows faster – without rebuilding the control bridge each time.



RESEARCH LABS & EDUCATORS

Teach language-grounded robotics in simulation before committing to hardware.



ROBOTICS STARTUPS

Build demos and pilots across different robot platforms without rewriting adapters.



ENTERPRISE INNOVATION INNOVATION TEAMS

Let operators use natural language within controlled, policy-approved workflows.

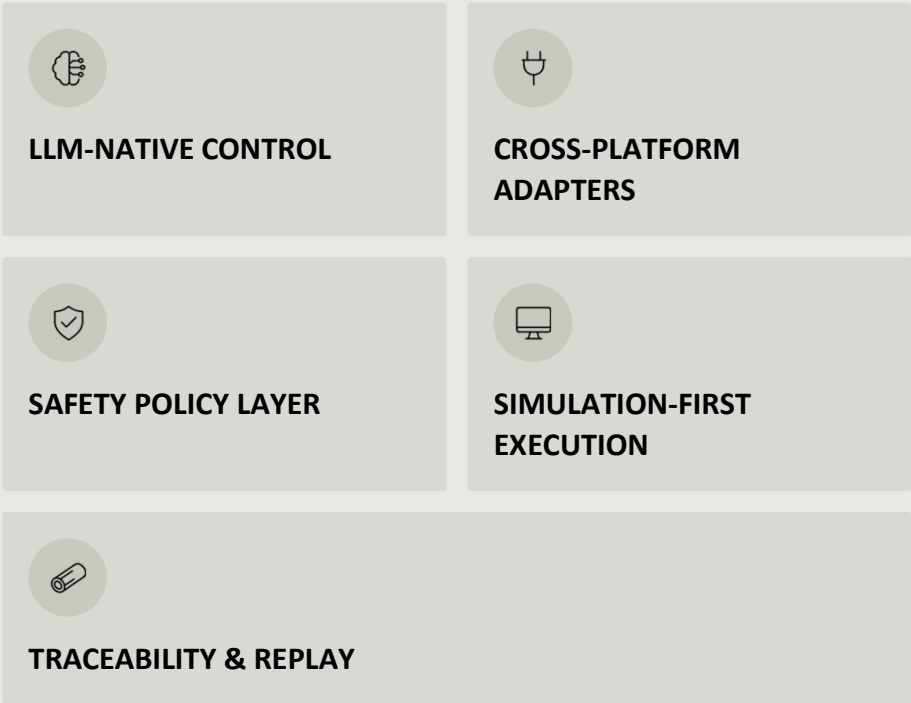
📄 **Beachhead:** Simulation-first developers and ROS 2 users.

THE MISSING LAYER BETWEEN LLMS AND ROBOT STACKS

EXISTING TOOLS SOLVE PARTS OF THE PROBLEM

Tool	Focus
ROSA / ROS-LLM	Natural language for ROS ecosystems
LeRobot / OpenVLA	Robot learning & model training
Viam	Robotics platform & machine APIs
Formant	Fleet observability & operations

ROBOT ABSTRACTION COMBINES ALL FIVE



OPEN-SOURCE ADOPTION, ENTERPRISE MONETIZATION



GO-TO-MARKET PLAYBOOK

01

LAUNCH

Hackathon demo + public GitHub repo to seed developer awareness.

02

GROW

Win ROS 2 and simulator developers through docs and community.

03

CONVERT

Active users upgrade to hosted telemetry and policy tools.

04

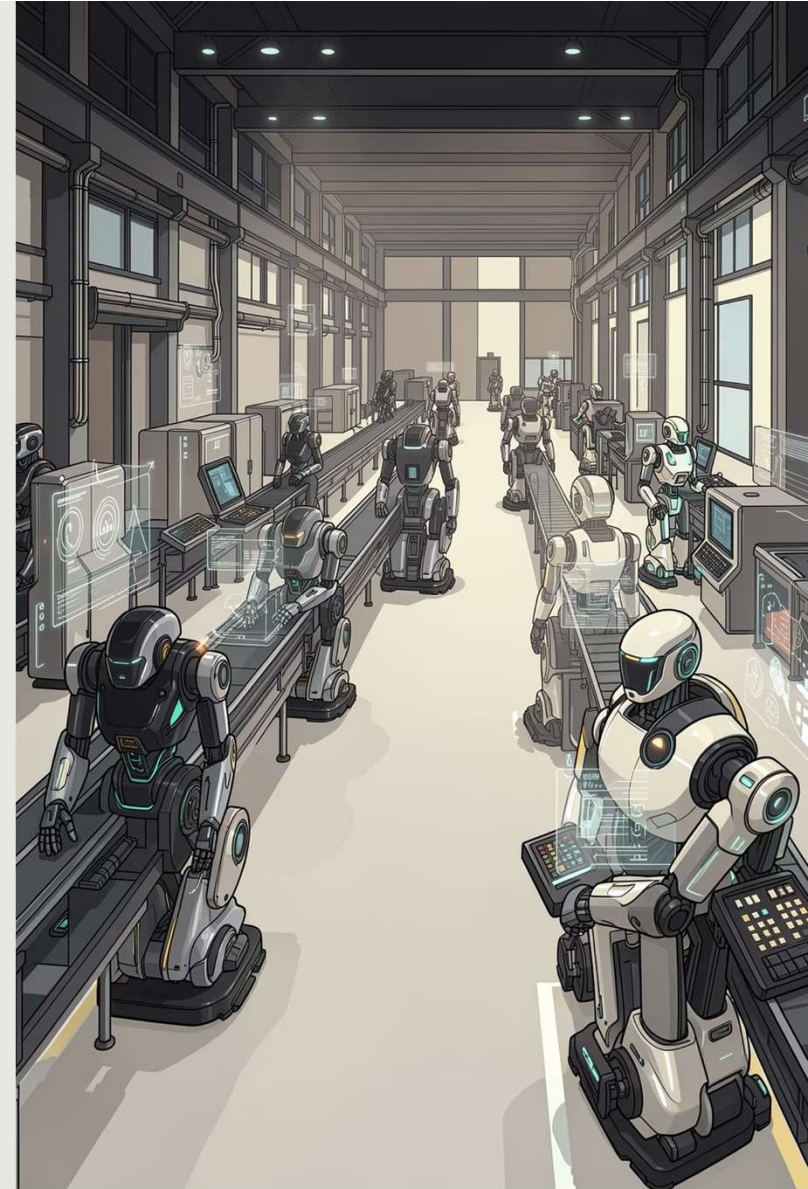
EXPAND

Land robotics startups and enterprise pilots at scale.

FROM HACKATHON MVP TO ROBOT CONTROL PLANE

- 1 — HACKATHON MVP**
Gemini orchestration, capability schema, safety gate, ROS 2 + simulator demo, trace logs.
- 2 — 30 DAYS**
TurtleBot3 profile, unsafe prompt test suite, public docs and quickstarts.
- 3 — 90 DAYS**
Hosted traces, multi-LLM routing, premium adapters, and policy templates.
- 4 — 12 MONTHS**
Enterprise control plane for heterogeneous robot fleets at scale.

Robots should be programmable by intent, governed by policy, and observable by default.



QUESTIONS?

LET'S BUILD THE FUTURE OF ROBOT CONTROL TOGETHER.

- Email: info@robotabstraction.com
- Explore the code: github.com/robot-abstraction
- Visit our website: robotabstraction.com

